

Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer

Phase 5: Technical Development & Implementation Report

LEAD DEVELOPER	DOCUMENT STATUS	UPDATE SET ID	DATE
Sr. ServiceNow Eng	Build Completed	US_PROC_AUTO_V1.2	Q4 2026

1. TECHNICAL DEVELOPMENT SCOPE OVERVIEW

Phase 5 details the concrete build artifacts, catalog items, client/server scripts, Flow Designer components, and REST Spoke integrations created during Sprints 1 through 3.

Implementation Principle

Adhere strictly to Low-Code/No-Code implementation using standard ServiceNow Utah/Vancouver platform components, isolating custom JavaScript to client-side GlideAjax script includes where necessary.

2. SERVICE CATALOG ITEM IMPLEMENTATION (CAT_ITEM_STANDARD_LAPTOP)

The Service Catalog item was constructed with high usability and dynamic client validation scripts:

Artifact Name	Type	Implementation Details
AutoFillRequesterData	Script Include	GlideAjax client-callable script returning JSON object containing user manager, cost center, and default location.
SetDynamicCatalogUI	Catalog Client Script	Triggered onLoad and onChange of requested_for to pre-populate form variables automatically.
EnforceTierJustification	Catalog UI Policy	Makes business_justification mandatory when selecting Developer Tier 2 or Executive Tier 3 laptops.

2.1 Server-Side Script Include Code Snippet

```
var AutoFillRequesterData = Class.create(); AutoFillRequesterData.prototype =
Object.extend(Object.prototype, { getUserDetails: function() { var userId =
this.getParameter('sysparm_user_id'); var gr = new GlideRecord('sys_user'); if (gr.get(userId)) {
return JSON.stringify({ manager: gr.getValue('manager'), cost_center: gr.getValue('cost_center'),
location: gr.getDisplayValue('location') }); } return null; }, type: 'AutoFillRequesterData' });
```

3. FLOW DESIGNER WORKFLOW EXECUTION LOGIC

The main workflow `flow_laptop_procurement` was configured in Flow Designer using native action steps and error evaluation handlers.

Step	Flow Action Component	Configuration Details
1.0	Trigger: Service Catalog	Fires when <code>sc_req_item</code> record is inserted for <code>cat_item_standard_laptop</code> .
2.0	Get Catalog Variables	Extracts <code>hardware_tier</code> , <code>requested_for</code> , <code>shipping_address</code> , and <code>cost_center</code> .
3.0	Make Decision	Evaluates Decision Table <code>dt_laptop_approval</code> . Returns <code>Auto_Approved</code> , <code>Manager_Approval</code> , or <code>Executive_Approval</code> .
4.0	Sub-flow: Request Approval	Sends actionable MS Teams card to approver. Waits for response or triggers 48h timeout escalation.
5.0	Sub-flow: Fulfill & Assign	Queries <code>alm_hardware</code> . Allocates local stock or calls REST Spoke <code>Vendor_Order_v2</code> .
6.0	Update Record & Notify	Updates RITM state to <code>Fulfillment Complete</code> and pushes shipment tracking code to user via Teams.

4. DECISION TABLE IMPLEMENTATION (DT_LAPTOP_APPROVAL)

Decision Tables decouple approval logic from workflow code, permitting non-technical business managers to adjust approval boundaries dynamically.

```
// Decision Table Condition Logic
IF (Hardware_Tier == "Tier 1 Standard") AND (Cost <= 1500 USD) -> Outcome: AUTO_APPROVE
IF (Hardware_Tier == "Tier 2 Developer") AND (Cost <= 2800 USD) -> Outcome: REQUIRE_MANAGER
IF (Hardware_Tier == "Tier 3 Executive") OR (Cost > 2800 USD) -> Outcome: REQUIRE_MANAGER_AND_VP
```

5. ERROR HANDLING & RETRY ARCHITECTURE

- REST Timeout Retry:** Automated 3-stage retry loop with Exponential Backoff (10s, 30s, 120s) for outbound vendor calls.
- Fallback Task Generation:** If vendor REST call fails after retries, flow automatically creates a high-priority manual task for the IT Procurement queue.

6. VENDOR REST SPOKE & CMDB AUTOMATION BUILD

Outbound integration with external suppliers was implemented using ServiceNow IntegrationHub REST Spoke actions.

Integration Attribute	Technical Specification
Credential Type	OAuth 2.0 Client Credentials stored in sys_auth_profile table.
HTTP Header	Content-Type: application/json Accept: application/json
JSON Response Handler	Script step parses vendor_order_id and estimated_ship_date into flow outputs.
CMDB Update Action	Calls Identification & Reconciliation Engine (IRE) to create or update CI in cmdb_ci_computer.

7. DEPLOYMENT UPDATE SET HIERARCHY

All development artifacts were grouped into audited ServiceNow Update Sets for migration into QA and Production:

Master Update Set Container: US_PROC_AUTO_MASTER_V1.0

- US_PROC_AUTO_CATALOG_V1 . 1 (Catalog Item, Variable Sets, Client Scripts)
- US_PROC_AUTO_FLOWS_V1 . 2 (Flow Designer, Sub-flows, Decision Tables)
- US_PROC_AUTO_INTEGRATION_V1 . 0 (REST Spoke, REST Credentials, Script Includes)

8. DEVELOPMENT SIGN-OFF & READINESS STATE

Deliverable Target	Build Status	Code Review	Date
Service Catalog Form	100% Completed	Passed	Q4 2026
Flow Designer Logic	100% Completed	Passed	Q4 2026
Vendor REST Integration	100% Completed	Passed	Q4 2026